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Inkjet printhead and method of manufacturing the same

Abstract

Disclosed are an inkjet printhead and a method of manufacturing the same, the inkjet printhead including: a first layer including an inlet formed to penetrate a substrate and introduce ink therein, and a plurality of membranes; a second layer disposed beneath the first layer, and including a manifold formed to penetrate a substrate and communicate with the inlet or recessed on a top of the substrate, and a plurality of nozzle channels formed to penetrate the substrate below the membrane and allow the ink transferred from the manifold to flow therein; a third layer disposed beneath the second layer, and including a plurality of nozzles formed in a substrate and communicating the plurality of nozzle channels; a piezoelectric actuator formed on the first layer formed with the membrane, and including a lower first electrode, a piezoelectric body on the first electrode, and a second electrode on the piezoelectric body; a first voltage controller configured to oscillate the membrane by applying a pulse voltage to the first electrode and the second electrode; a third electrode disposed beneath the third layer, formed around each nozzle, and surrounded with an insulator; and a second voltage controller configured to discharge droplets of the ink based on induced electric force by applying voltage to the third electrode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/656,401, filed Mar. 24, 2022; which claims priority from

Korean Patent Application No. 10-2022-0029786, filed Mar. 10, 2022; this application also claims priority from Korean Patent Application No. 10-2022-0169557, filed Dec. 7, 2022, the disclosures of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(a) Field of the Invention

(1) The disclosure relates to an inkjet printhead and a method of manufacturing the same, and more particularly to an inkjet printhead and a method of manufacturing the same, in which a multi-nozzle is used to discharge ink in a hybrid way based on combination of a piezoelectric method and an induced electrohydrodynamic method that discharges a charged solution by electrostatic force induced on a liquid surface at the end of the nozzle.

(b) Description of the Related Art

(2) In general, an inkjet printhead refers to a device that prints an image with predetermined colors on a surface of a printing medium by discharging fine droplets of ink to a desired position on the printing medium. The applications of the inkjet printhead have recently expanded to various fields such as a liquid crystal display (LCD), an organic light emitting device (OLED) and the like flat panel display fields; electronic (E)-paper and the like flexible display fields; metal wiring and the like printed electronics field; bio fields; and so on.

(3) Drop-on-demand (DOD) inkjet printhead is classified according to discharging methods into a piezoelectric inkjet printhead that discharges ink by pressure waves based on transformation of a piezoelectric body, and an electrohydrodynamic inkjet printhead that discharges ink by electrostatic force.

(4) In the piezoelectric inkjet printhead, a piezoelectric body vibrates a membrane to apply pressure to a chamber containing ink, thereby discharging the ink. In general, droplets are discharged when pressure high enough to overcome the surface tension and viscosity of the ink on the surface of a nozzle is applied, and pressure applied additionally is required to be high enough to accelerate the discharged droplets to a speed at which these droplets can be accurately settled on a printing medium. To discharge a droplet of several picoliters or less, the piezoelectric inkjet printhead needs to reduce transformation energy in a pressure chamber. When the transformation energy in the pressure chamber is reduced, discharging energy per unit volume of the discharged droplets is also reduced, and therefore a discharging speed of the droplets is decreased. However, when the discharging speed of the droplets is decreased, a problem arises in that the droplets are not accurately discharged to desired positions.

(5) The piezoelectric inkjet printhead is advantageous in that it is easy to control a printing job, and there are no restrictions on the types of ink because the discharging energy is based on mechanical transformation. However, the piezoelectric inkjet printhead has difficulty in discharging ultrafine droplets of several picoliters or less, and has a limitation in that the discharge of only ink having a viscosity of about 10 cPs is possible but the discharge of ink having a high viscosity is not possible. Further, it is difficult to discharge big droplets of 80 picoliters or more due to the limitations on the discharging energy. In particular, the piezoelectric inkjet printhead has a limitation even though volume uniformity of discharged droplets between a plurality of nozzles is very important for applications to the processes of the printed electronics such as a display, etc. unlike the existing graphic printing.

(6) On the other hand, the electrohydrodynamic inkjet printhead provides the discharging energy by applying electrostatic force to a liquid surface of ink formed at the end of a nozzle, and is therefore advantageous in that the discharge of ultrafine droplets of not more than several picoliters or femtoliters is possible and the discharge of ink droplets having a high viscosity of about 1,000 cPs is possible. Besides, it is also possible to discharge big droplets of 80 picoliters or more. The electrohydrodynamic inkjet printhead is advantageous for precise printing because a driving method is simple and the directionality of discharged ink droplets is excellent due to control based

on distribution of an electric field formed on the nozzle. However, the electrohydrodynamic inkjet printhead has difficulty in cleaning the nozzle, which needs to be performed during a process, because the nozzle has a protruding structure for the concentration of the electric field. An inkjet head generally secures the stability of droplet jetting by cleaning the nozzle, which is contaminated with the ink, through wiping or the like process. However, it is not easy to apply such a cleaning process to the protruding nozzle. When the droplets are discharged by an electrohydrodynamic inkjet method, a voltage of several hundred V to several KV is applied. Further, a voltage controller having a very high slew rate is required to discharge the droplets by a DOD method. Therefore, there is a limit to discharging the droplets at a high frequency to secure sufficient discharging energy. It is known that the electrohydrodynamic inkjet method has the maximum jetting frequency of about 1 kHz. However, this jetting frequency is very low as compared to the high jetting frequency of 100 kHz the foregoing piezoelectric inkjet method has.

(7) Accordingly, there has been proposed a method of combining the foregoing two methods to minimize the shortcomings and utilize the advantages.

(8) In terms of combining the two methods, an electrohydrodynamic high-voltage electrode is disposed inside a nozzle, an ink chamber or an ink inlet. However, contact between the ink and the electrode causes an electro-chemical reaction that generates heat, and therefore problems arise in that the ink is denaturalized, bubbles are generated, the nozzle is clogged, etc. Further, there have been many technical difficulties such as difficulties in processing a membrane structure of the piezoelectric body, an electrode for electrohydrodynamic inkjet driving, etc.

(9) In addition, there has been a problem in that printing quality is poor because the droplets discharged through the multi-nozzle are not uniform in size.

DOCUMENT OF RELATED ART

Patent Document

(10) (Patent document 1) Korean Patent No. 10-0917279

SUMMARY OF THE INVENTION

(11) Accordingly, the disclosure is conceived to solve the foregoing problems, and an aspect of the disclosure is to provide an inkjet printhead and a method of manufacturing the same, in which, as an inkjet apparatus having a layered multi-nozzle, a pressure wave based on oscillation of a piezoelectric body controls a liquid surface at the end of the nozzle, and a droplet is discharged by applying electrostatic force based on an induced electric field to the liquid surface, thereby overcoming viscosity and surface tension to discharge ultrafine droplets or large droplets based on combination between mechanical driving force of the piezoelectric body and the electrostatic force, and improving size uniformity of droplets between the nozzles based control of the electrostatic force for each nozzle.

(12) The problems to be solved by the disclosure are not limited to those mentioned above, and other unmentioned problems will become apparent to a person skilled in the art by the following descriptions.

(13) In accordance with an embodiment of the disclosure, there is provided an inkjet printhead including: a first layer including an inlet formed to penetrate a substrate and introduce ink therein, and a plurality of membranes; a second layer disposed beneath the first layer, and including a manifold formed to penetrate a substrate and communicate with the inlet or recessed on a top of the substrate, and a plurality of nozzle channels formed to penetrate the substrate below the membrane and allow the ink transferred from the manifold to flow therein; a third layer disposed beneath the second layer, and including a plurality of nozzles formed in a substrate and communicating the plurality of nozzle channels; a piezoelectric actuator formed on the first layer formed with the membrane, and including a lower first electrode, a piezoelectric body on the first electrode, and a second electrode on the piezoelectric body; a first voltage controller configured to oscillate the membrane by applying a pulse voltage to the first electrode and the second electrode; a third electrode disposed beneath the third layer, formed around each nozzle, and surrounded with an

insulator; and a second voltage controller configured to discharge droplets of the ink based on induced electric force by applying voltage to the third electrode.

(14) Here, the first layer may include a plurality of chambers recessed inward from a bottom thereof to store the ink and formed with the membrane above a top thereof.

(15) Here, the chamber may include a first side communicating with the manifold, and a second side communicating with the nozzle channel.

(16) Here, the inkjet printhead may further include a hydrophobic coating layer disposed beneath an insulating layer formed the insulator and coated with a hydrophobic material.

(17) Here, the hydrophobic coating layer may be coated from an end of the nozzle to an inside of the nozzle.

(18) Here, voltage applied by the first voltage controller may be synchronized with voltage applied by the second voltage controller.

(19) Here, the second voltage controller may apply voltage to the third electrode to discharge a droplet when a meniscus is formed at an end of the nozzle as the first voltage controller applies a pulse voltage between the first electrode and the second to make the piezoelectric actuator oscillate the membrane, and the second voltage controller may apply a voltage having an opposite polarity to a discharging voltage or applies a voltage of 0V after the discharged droplet passes the third electrode.

(20) Here, the inkjet printhead may further include a fourth electrode disposed beneath the third electrode, surrounded with an insulating layer, and disposed encompassing an outlet having a larger diameter than an opening of the nozzle; and a third voltage controller configured to apply voltage to the fourth electrode.

(21) Here, voltage applied by the second voltage controller may be synchronized with voltage applied by the third voltage controller.

(22) Here, a horizontal distance between the fourth electrode and the nozzle may be longer than a horizontal distance between the third electrode and the nozzle.

(23) Here, the plurality of nozzles may be arranged in a matrix, and the third electrodes arranged in one of a row direction and a column direction may be electrically connected to simultaneously receive voltage from the second voltage controller, and the fourth electrodes arranged in the other one of the row direction and the column direction may be electrically connected to simultaneously receive voltage from the third voltage controller.

(24) Here, the inkjet printhead may further include a fifth electrode disposed between the first layer and the second layer and surrounded with an insulating layer; and a fourth voltage controller configured to apply voltage to the fifth electrode.

(25) Here, the first voltage controller may apply the same pulse voltage to the piezoelectric actuators respectively corresponding to the membranes, and the second voltage controller may apply different voltages to the third electrodes according to the nozzles to make droplets discharged from the nozzles be uniform in size.

(26) Here, the fourth voltage controller may apply voltage, which has an opposite polarity to the voltage of the second voltage controller, to the fifth electrode, or may serve as the ground.

(27) Here, the pulse voltage applied by the first voltage controller may be synchronized with the pulse voltage applied by the fourth voltage controller, so that electrostatic force based on potential difference between the first electrode and the fifth electrode can reinforce the oscillation of the membrane.

(28) Here, the third electrodes may be formed as a single body for a plurality of nozzles.

(29) Further, in accordance with an embodiment of the disclosure, there is provided a method of manufacturing an inkjet printhead, including: manufacturing a first layer including an inlet formed to penetrate a substrate and introduce ink therein, and a plurality of membranes on which a first electrode of a piezoelectric actuator is formed, a second layer including a manifold formed to penetrate a substrate and communicate with the inlet or recessed on a top of the substrate, and a

plurality of nozzle channels formed to penetrate the substrate below the membrane and allow the ink transferred from the manifold to flow therein, and a third layer including a plurality of nozzles formed in a substrate and communicating the plurality of nozzle channels, and a third electrode disposed beneath the third layer, formed around each nozzle, and surrounded with an insulator; joining the first layer, the second layer, and the third layer in sequence; and forming a piezoelectric body and a second electrode for a piezoelectric actuator on the first electrode.

(30) Here, the first layer may include a plurality of chambers recessed inward from a bottom thereof to store the ink and formed with the membrane above a top thereof.

(31) Here, the method may further include coating a hydrophobic material on a lower layer of the insulator when the third layer is manufactured.

(32) Here, a hydrophobic coating layer coated with the hydrophobic material may be coated from an end of the nozzle to an inside of the nozzle.

(33) Here, the method may further include forming a fourth electrode, disposed beneath the third electrode, surrounded with an insulator, and having a larger diameter than the third electrode when the third layer is manufactured.

(34) Here, the method may further include forming a fifth electrode disposed on the second layer and surrounded with an insulator when the third layer is manufactured.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of the exemplary embodiments, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a cross-sectional view of an inkjet printhead according to a first embodiment of the disclosure;

(3) FIG. 2 is a view showing an operation of a membrane based on a piezoelectric body of FIG. 1;

(4) FIGS. 3A and 3B are views showing synchronous signals of a first voltage controller and a second voltage controller according to an embodiment of the disclosure;

(5) FIG. 4 illustrates simulation results of meniscus shapes according to horizontal distances between a nozzle and a third electrode;

(6) FIGS. 5A and 5B illustrate simulation results of electric field distributions according to vertical distances between the nozzle and the third electrode;

(7) FIG. 6 is a cross-sectional view of an inkjet printhead according to a second embodiment of the disclosure;

(8) FIG. 7 illustrates a voltage applying structure of a third electrode and a fourth electrode, and FIG. 8 illustrates a voltage applying operation when ink is discharged from only some nozzles among the nozzles;

(9) FIG. 9 is a cross-sectional view of an inkjet printhead according to a third embodiment of the disclosure;

(10) FIG. 10 is a view showing an operation of a membrane based on a piezoelectric body and a fifth electrode of FIG. 9;

(11) FIG. 11 is a cross-sectional view of an inkjet printhead according to a fourth embodiment of the disclosure;

(12) FIG. 12 is a flowchart showing a method of manufacturing an inkjet printhead according to an embodiment of the disclosure;

(13) FIG. 13 is a view showing a method of manufacturing a chamber layer;

(14) FIG. 14 is a view showing a method of manufacturing a channel layer; and

(15) FIG. 15 is a view showing a method of manufacturing a nozzle layer.

DETAILED DESCRIPTION

(16) Specific features of embodiments are involved in the detailed description and the accompanying drawings.

(17) The merits and features of the disclosure, and methods of achieving them will become apparent with reference to the embodiments described below in detail and the accompanying drawings. However, the disclosure is not limited to the embodiments set forth herein, but may be implemented in various forms. The following embodiments are provided in order to fully describe the disclosure and enable those skilled in the art, to which the disclosure pertains, to understand the disclosure, the scope of which is defined in the appended claims. Like numerals refer to like elements throughout.

(18) Below, embodiments of an inkjet printhead and a method of manufacturing the same according to the disclosure will be described with reference to the accompanying drawings.

(19) FIG. 1 is a cross-sectional view of an inkjet printhead according to a first embodiment of the disclosure, and FIG. 2 is a view showing an operation of a membrane based on a piezoelectric body of FIG. 1.

(20) An inkjet printhead according to the first embodiment of the disclosure may include a first layer **110**, a second layer **120**, a third layer **130**, a piezoelectric actuator **140**, a first voltage controller **145**, an electrode (i.e., a third electrode **152**) for electrohydrodynamic jetting, and a second voltage controller **155**.

(21) The first layer (hereinafter, referred to as a chamber layer) **110** may be formed with an inlet **111** into which ink is introduced from the outside, and a plurality of membranes **115** which oscillates by the piezoelectric actuator **140**. The inlet **111** is formed to vertically penetrate a substrate, which forming the chamber layer **110**, at a predetermined position, and the substrate is formed with the thin membrane **115** having a predetermined thickness. The inlet **111** refers to an opening through which ink stored in an ink storage tank (not shown) flows into the inside of the printhead.

(22) As shown therein, a chamber **112** may be formed as recessed inward on the bottom of the substrate, and filled with the ink supplied through the inlet **111**, and the membrane **115** may be formed above the chamber **112**. Although not shown, a plurality of chambers **112** are spaced apart from each other in a direction perpendicular to FIG. 1, and thus the chamber **112** and the membrane **115** are formed corresponding to each of a plurality of nozzles **131**.

(23) On the membrane **115** of the chamber layer **110**, a first electrode **141**, a piezoelectric body **142** on the first electrode **141**, and a second electrode **143** on the piezoelectric body **142**, which form the piezoelectric actuator **140**, may be stacked in sequence. Further, a pulse voltage may be applied between the first electrode **141** and the second electrode **143** through the first voltage controller **145**.

(24) The first electrode **141** serves as a common electrode, and the second electrode **143** serves as a driving electrode for applying the pulse voltage to the piezoelectric body **142**. Therefore, a plurality of second electrodes **143** may be individually disposed with regard to the plurality of nozzles **131**. The piezoelectric body **142** may contain a predetermined piezoelectric material, for example, a piezoelectric transducer (PZT).

(25) In this case, the second electrode **143** on the piezoelectric body **142** is formed as the common electrode, and a plurality of first electrodes **141** beneath the piezoelectric body **142** are disposed corresponding to the plurality of nozzles **131**, thereby applying the pulse voltage.

(26) When the first voltage controller **145** applies the pulse voltage between the two electrodes **141** and **143** forming the piezoelectric actuator **140**, the membrane **115** may be transformed up and down by the operation of the piezoelectric body **142**. In this case, the membrane **115** serves as a vibrating plate that generates pressure waves in the chamber **112** based on the transformation of the membrane **115**.

(27) The voltage applied by the first voltage controller **145** may be a pulse voltage in which a

positive voltage or a negative voltage is periodically generated having a predetermined amplitude. Further, the voltage applied by the first voltage controller **145** may be a pulse voltage in which a positive voltage and a negative voltage are periodically generated having a predetermined amplitude. In addition, the voltage applied by the first voltage controller **145** may be an alternative current (AC) voltage having a predetermined waveform such as a sine wave, a triangular wave, and so on.

(28) The second layer (hereinafter, referred to as a channel layer) **120** is disposed beneath the chamber layer **110**, and may include a manifold **121** formed penetrating a substrate having a predetermined thickness and communicating with the inlet **111**, and a plurality of nozzle channels **122** formed penetrating the substrate and allowing the ink to flow from a first side of the manifold **121**. As shown therein, a restrictor **113** is formed below a first side of the chamber **112** to communicate with the manifold **121** and communicates with the nozzle channel **122** below a second side of the chamber **112**.

(29) Therefore, the ink flowing from the outside into the chamber layer **110** through the inlet **111** is stored in the space of the manifold **121** formed together with the top of the nozzle layer **130**, and the ink stored in the manifold **121** is transferred to each chamber **112** via the restrictor **113** and then transferred to the nozzle channel **122**. The nozzle channel **122** is disposed between the chamber **112** of the chamber layer **110** and the nozzle **131** of the nozzle layer **130**.

(30) The restrictor **113** serves to restrict the pressure waves traveling toward the nozzle and traveling toward the manifold **121** due to the transformation of the piezoelectric body **142**. To this end, the cross-sectional area of the restrictor **113** may be equal to or smaller than the cross-sectional area of the nozzle channel **122**.

(31) In the accompanying drawings, the manifold **121** penetrates the substrate and forms a space, in which the ink is stored, together with the top of the third layer **130**. Alternatively, the manifold may be recessed on the top of the substrate at a predetermined depth without penetrating the substrate.

(32) The third layer (hereinafter, referred to as the nozzle layer **130**) is disposed beneath the channel layer **120**, and includes the plurality of nozzles **131** formed on a substrate having a predetermined thickness and communicating with the nozzle channel **122**. Through the nozzle **131**, the ink may be discharged forming a droplet toward a printing medium (not shown) put under the nozzle **131**.

(33) As shown therein, the nozzle **131** is formed penetrating the nozzle layer **130**. In the nozzle layer **130**, the nozzle **131** having a relatively small diameter may be formed in an upper portion, and an outlet **154** having a relatively large diameter may be formed in a lower portion. Therefore, openings are formed with stepped increase in diameter downward while penetrating the nozzle layer **130**. With this structure, a meniscus may be formed at not the end of the outlet **154** but the end of the nozzle **131** due to the operation of the piezoelectric actuator **140**.

(34) On the bottom of the nozzle layer **130**, a plurality of third electrodes **152** may be formed being surrounded with an insulating layer **151**. The third electrode **152** may be formed in every nozzle **131**. In this case, the third electrode **152** may have various shapes such as a circular shape, a horseshoe shape, a quadrangular shape, a diamond shape, etc. Preferably, the third electrode **152** may be shaped to surround all or some of the nozzles **131**.

(35) Further, the third electrodes **152** formed in the nozzles **131** may be formed not separately but integrally. In other words, the third electrodes **152** are formed for the nozzles **131**, and a connecting structure is formed for connection between adjacent third electrodes **152**, thereby forming the third electrodes **152** as a single body. Alternatively, the third electrodes **152** may be individually formed being respectively separated from the nozzles **131**.

(36) The third electrode **152** surrounded with the insulating layer **151** may be disposed beneath the bottom of the nozzle layer **130** in such a way that a first insulating layer is formed beneath the bottom of the nozzle layer **130**, the third electrode **152** is formed beneath the first insulating layer, and a second insulating layer is formed beneath the third electrode **152**. As shown therein, when the

nozzle layer **130** is manufactured with a glass wafer, the third electrode **152** may be disposed directly under the nozzle layer **130** because the glass wafer is an insulating material, and the insulating layer **151** may be formed beneath the third electrode **152**.

(37) In this case, the insulating layer may be formed not only beneath the bottom of the nozzle layer **130** but also on the inner surfaces of the outlet **154** and the nozzle **131**.

(38) The third electrode **152** may be connected to the second voltage controller **155** and receive high voltage.

(39) In addition, a hydrophobic coating layer **160** coated with a hydrophobic material may be formed beneath the insulating layer **151**. The hydrophobic coating layer **160** may inhibit the bottom of the printhead, which is likely to come into contact with liquid, from contamination. In this case, the hydrophobic coating layer **160** may also be formed on the inner surface of the outlet **154**.

(40) Further, as shown therein, the hydrophobic coating layer **160** may be coated up to a predetermined depth of the nozzle **131** inward from the end. When the hydrophobic coating layer **160** is formed inward from the end of the nozzle **131**, a droplet can be formed inside the nozzle **131**, thereby increasing the printing stability and showing improvement in terms of maintenance.

(41) In this case, the chamber layer **110**, the channel layer **120**, and the nozzle layer **130** may be made of silicon wafers. Alternatively, the channel layer **120** and the nozzle layer **130** may also be made of glass wafers.

(42) As shown in FIG. 2, when the first voltage controller **145** applies the pulse voltage between the first electrode **141** and the second electrode **143** of the piezoelectric actuator **140**, the piezoelectric body **142** operates to oscillate the membrane **115** up and down. The oscillation of the membrane **115** transfers a pressure wave to the nozzle channel **122**, so that a concave or convex meniscus can be formed at the end of the nozzle **131** by negative pressure or positive pressure of the pressure wave.

(43) When the second voltage controller **155** applies high voltage to the third electrode **152** under the condition that the piezoelectric actuator **140** forms the convex meniscus at the end of the nozzle **131** with the positive pressure of the pressure wave, a droplet may be discharged from the meniscus by electrostatic force generated by induced charges. When the membrane **115** is moved up by the piezoelectric actuator **140** and the chamber **112** is increased in volume, the change in pressure causes the ink stored in the manifold **121** to flow into the chamber **112** through the restrictor **113**.

(44) Voltage applied to the third electrode **152** may be a positive or negative voltage having a predetermined amplitude. Alternatively, the voltage may be applied in the form of pulses, and thus synchronized with the pressure waves of the piezoelectric body **142**. Alternatively, an AC voltage alternating between a positive voltage and a negative voltage may be applied, so that electric charges for discharging a droplet can be stabilized at the end of the nozzle **131**.

(45) Alternatively, a constant bias voltage may be applied to the third electrode **152**. In an electrohydrodynamic inkjet apparatus that uses the electrostatic force to discharge a droplet, there is an on-set voltage with which the droplet starts being discharged with respect to the nozzle **131** having a specific size and the ink having a specific dielectric constant. The bias voltage may be equal to or lower than the on-set voltage.

(46) As shown in FIGS. 3A and 3B, the second voltage controller **155** may apply high voltage to the third electrode **152** from time when the meniscus at the end of the nozzle **131** starts becoming convex by positive pressure as the first voltage controller **145** applies a voltage pulse for driving the piezoelectric body **142** and the pressure wave is transferred to the end of the nozzle **131**. In other words, when the surface of liquid at the end of the nozzle **131** forms a convex meniscus, electrohydrodynamic discharge may actually start. The discharge in a given nozzle **131** may begin as an electric field is applied to a surrounding area of the nozzle **131** by the third electrode **152**. Preferably, the electric field is generated by difference in electric potential between the ink grounded or charged with specific polarity and the third electrode **152**. The ink does not need to be grounded, but may be grounded to make the ink including charged particles or the like have

uniform potential difference.

(47) Due to an electric field generated when high voltage is applied to the third electrode **152**, the surface of the ink forming the convex meniscus may be charged. By interaction between the charged ink meniscus and the electric field, force is generated to change the shape of the meniscus. When the electric field is strengthened to make the convex meniscus become more convex by increasing the potential difference between the third electrode **152** and the ink until stress electrically induced on the meniscus overcomes the surface tension of the ink, the discharge of the droplet may occur.

(48) In this case, although not shown, a separate electrode for generating the electric field together with the third electrode **152** may be additionally provided under a printing medium.

(49) As shown in FIG. 3A, the voltage applied to the third electrode **152** may be controlled to fall to 0V after a droplet is detached from the meniscus formed at the end of the nozzle **131** by the voltage V applied to the third electrode **152** and passes the third electrode **152**. Alternatively, as shown in FIG. 3B, the applied voltage may be controlled to have the opposite polarity ($-V'$) after a droplet is detached from the meniscus formed at the end of the nozzle **131** by the voltage V applied to the third electrode **152** and passes the third electrode **152**.

(50) If voltage having the same polarity is continuously applied to the third electrode **152** even after the droplet passes the third electrode **152**, force of attraction acts to pull the droplet in the opposite direction to the discharging direction, thereby making the move of droplets unstable and causing wetting. Thus, according to the disclosure, the voltage applied to the third electrode **152** is controlled to fall to 0V or to have the opposite polarity after the droplet passes the third electrode **152**, thereby solving the foregoing problems. When the voltage having the opposite polarity is applied after the droplet passes the third electrode **152**, force of repulsion may act on the droplet by the third electrode **152**.

(51) Therefore, according to the disclosure, as shown in FIGS. 3A and 3B, the first voltage controller **145** and the second voltage controller **155** may apply synchronized voltages to the two electrodes **141** and **142** of the piezoelectric actuator **140** and the third electrode **152**, respectively.

(52) In more detail with respect to the synchronization, when defined that the pulse time applied from the first voltage controller **145** is t_1 , the waiting time until the next pulse voltage is applied is t_6 , the time between when the pulse voltage is applied from the first voltage controller **145** and the voltage is applied from the second voltage controller **155** is t_2 , the pulse time applied from the second voltage controller **155** is t_3 (FIG. 3A), and when an AC voltage is applied from the second voltage controller **155** (FIG. 3B) a pulse time of one polarity is t_4 , and a pulse time of the opposite polarity is t_5 ,

(53) Since the basic meniscus is formed by the voltage applied from the first voltage controller **145** and the droplet is jetted by the voltage applied from the second voltage controller **155**, it must be synchronized with the following conditions. (1) $t_1+t_6 \geq t_2+t_3$ or $t_1+t_6 \geq t_2+t_4+t_5$ (2) $t_1 \geq t_2$ (3) t_3 or t_4 is applied until the droplet is formed (4) $t_4 \leq t_3$, t_5 accelerates the droplet

(54) According to the disclosure, the third electrode **152** is formed below the nozzle **131** and separated from the nozzle **131** by the insulating layer **151**. Therefore, the insulating layer **151**, by which the third electrode **152** and the ink can be separated, inhibits an oxidation-reduction reaction between the ink and the third electrode **152** when high voltage is applied to the third electrode **152**, and solves problems such as heat generation, ink denaturalization, bubble generation, nozzle clogging, etc. due to the oxidation-reduction reaction.

(55) Although the third electrode **152** separated by the insulating layer **151** does not directly apply electric charges to a liquid surface of the nozzle **131**, it is possible to concentrate induced charges on the liquid surface of the nozzle. Therefore, according to the disclosure, the droplets are discharged toward the printing medium by electrostatic force generated by induced charges when high voltage is applied to the third electrode **152**.

(56) According to the disclosure, the nozzle **131** is provided as a multi-nozzle. The droplets

discharged through the multi-nozzle may be different in size from each other. A conventional inkjet printhead using only the driving force of the piezoelectric body is capable of discharging droplets at an ultrahigh frequency of 100 kHz, but has a problem in that deviation in size between the droplets from the multi-nozzle is large. Accordingly, a drive per nozzle (DPN) inkjet printhead has been developed to individually control the piezoelectric body for each nozzle, but problems arise in that there is difficulty in the individual control, and it is difficult to optimize the shape of pulses for controlling the piezoelectric body for each nozzle. On the other hand, according to the disclosure, the first voltage controller **145** controls voltage to be equally applied to the piezoelectric actuators **140** respectively corresponding to the nozzles **131**, and the second voltage controller **155** controls the voltage applied to the third electrode **152** to be different in level and frequency according to the nozzles **131**, thereby controlling the droplets discharged from the multi-nozzle to be uniform in size. It is much simpler to control the size of droplet based on the level of high voltage applied to the third electrode **152** than to control each pulse voltage for the piezoelectric actuator **140**.

(57) According to the disclosure, the diameter of the droplet may be considerably smaller than the diameter of the nozzle **131** from which the droplet is discharged. In general, the convex meniscus has a diameter given as the outer diameter of the entrance of the nozzle **131**. According to the disclosure, the diameter of the droplet may be adjusted within a range from about 1/20 of the diameter of the nozzle **131** to the same diameter as the nozzle **131**, by adjusting the voltage applied to the nozzle **131**. However, in principle, a droplet smaller than 1/20 of the diameter of the nozzle **131** may also be producible. Depending on change in voltage, a droplet is varied in size. The droplet is increased in diameter as voltage increases between the minimum extraction voltage at which the applied voltage is needed for discharging the droplet and an extraction voltage approximately twice the minimum extraction voltage, and a diameter change rate of the droplet is decreased as the extraction voltage increases.

(58) The use of the nozzle **131** having a large diameter to produce much smaller droplets has various advantages. First, it is much easier to manufacture the nozzle **131** having a large diameter by the existing manufacturing method. Low resolution requirements may have a significant effect on costs and time required for manufacturing the printhead. Next, it is much faster to print a structure on a predetermined area of the substrate at a given resolution through the nozzle **131** having a large diameter. Further, when the discharged droplet has a diameter considerably smaller than the diameter of the nozzle **131**, the discharging amount and the droplet size are much less affected by change in voltage. Accordingly, the droplets deposited by the nozzles **131** can have the same diameter and be discharged at the same frequency even though there is a slight difference in diameter between the manufactured nozzles **131**. Besides, the nozzles are free from being clogged with contaminants or ink attached to and dried on the printhead. In addition, it is easier to clean the nozzle **131** having a larger diameter.

(59) As shown in FIG. 1, the third electrode **152** may be formed encompassing the nozzle **131** below the end of the nozzle **131**. FIG. 4 shows simulation results of effects on a liquid surface of a meniscus according to horizontal distances b between the third electrode **152** and the nozzle **131**. When the horizontal distance b between the third electrode **152** and the nozzle **131** gets shorter into 50 μm , 30 μm and 10 μm , wetting effects are increased as the third electrode **152** more attracts the edges of the meniscus. Therefore, the horizontal distance b between the third electrode **152** and the nozzle **131** needs to be designed to be greater than or equal to a predetermined minimum distance in order to inhibit the wetting effects due to the force of attraction of the third electrode **152**. In this case, the minimum distance may increase as the diameter of the nozzle **131** increases.

(60) FIGS. 5A and 5B show simulation results of effects on distribution of an electric field according to vertical distances a between the third electrode **152** and the end of the nozzle **131**. As the vertical distance a between the third electrode **152** and the nozzle **131** gets shorter, the droplet discharging energy increases due to the effects of the electric field but the concentration of the electric field decreases (see FIG. 5A). On the other hand, as the vertical distance a between the

third electrode **152** and the nozzle **131** gets longer, the droplet discharging energy decreases but the concentration of the electric field increases (see FIG. 5B). When the concentration of the electric field decreases, the droplet may become unstable. Accordingly, the optimum vertical distance a between the third electrode **152** and the nozzle **131** is designed by taking the discharging energy of the droplet and the concentration of the electric field into account.

(61) Further, the vertical distance a between the third electrode **152** and the nozzle **131** is much shorter than the distance between the printhead and the printing medium. Therefore, a strong electric field is locally formed around the nozzle **131** by not the distance between the printhead and the printing medium but the third electrode **152**.

(62) Below, an inkjet printhead according to a second embodiment of the disclosure will be described with reference to FIGS. 6 to 8.

(63) FIG. 6 is a cross-sectional view of an inkjet printhead according to a second embodiment of the disclosure, FIG. 7 illustrates a voltage applying structure of a third electrode and a fourth electrode, and FIG. 8 illustrates a voltage applying operation when ink is discharged from only some nozzles among the nozzles.

(64) In terms of describing the second embodiment, descriptions will be made focusing on differences from the foregoing embodiment.

(65) Even in this embodiment, the inkjet printhead has a three-layered structure of the chamber layer **110**, the channel layer **120** and the nozzle layer **130**.

(66) As described above, the third electrode **152** coated with the insulating layer **151** is disposed beneath the bottom of the nozzle layer **130**. The third electrode **152** and a fourth electrode **153** are all disposed inside the insulating layer **151** beneath the nozzle layer **130**, in which the fourth electrode **153** is disposed below and spaced apart from the third electrode **152**, and the horizontal distance between the fourth electrode **153** and the nozzle **131** is designed to be longer than the horizontal distance between the third electrode **152** and the nozzle **131**.

(67) The hydrophobic coating layer **160** coated with a hydrophobic material may be formed beneath the insulating layer **151** surrounding the third electrode **152** and the fourth electrode **153**.

(68) A third voltage controller **157** applies voltage to the fourth electrode **153**.

(69) When a droplet is formed and discharged by induced electrostatic force of the third electrode **152** and the pressure wave caused by the piezoelectric body **142** as the first voltage controller **145** applies voltage to the electrodes **141** and **143** of the piezoelectric actuator **140** and the second voltage controller **155** applies voltage to the third electrode **152**, the third voltage controller **157** applies voltage for forming the electrostatic force to the fourth electrode **153**, thereby narrowing a discharging angle of droplets and reinforcing a discharging speed. In this case, the voltage applied to the fourth electrode **153** is synchronized to have the same phase as the voltage applied to the third electrode **152**.

(70) The vertical distance between the third electrode **152** and the end of the nozzle **131** is much shorter than the distance between the printhead and the printing medium. Therefore, a strong electric field is locally formed around the nozzle **131** by not the distance between the printhead and the printing medium but the third electrode **152**.

(71) By limiting the width of the third electrode **152**, it is possible to limit the area of the strong electric field to a place where the individual nozzles **131** is positioned, in a transverse direction, and narrow the distance between the nozzles **131**. However, when the distance from the neighboring third electrodes **152** becomes closer, the electric field of the neighboring nozzle **131** may be affected. Thus, the fourth electrode **153** is useful for inhibiting the non-uniformity and transverse deflection of the electric field as affected by the electric field from the neighboring third electrode **152**.

(72) The third electrode **152** needs to have the same polarity and a preferably higher amplitude as compared to extraction potential applied to liquid potential while the droplets are discharged, with regard to the droplets discharged as accelerated in an intended direction, i.e., a discharging

direction. The accelerating electric field may be accurately orthogonal to the surface of the printing medium or the printhead, so that the droplet can be accelerated in a direction perpendicular to the surface of the printhead without being deflected in the transverse direction.

(73) Voltage applied to the fourth electrode **153** may be a direct current (DC) voltage, and may preferably be given in the form of a continuous signal having a constant or variable amplitude. Alternatively, the applied voltage may be an AC voltage, and may preferably be given in the form of a periodic function having a 20 Hz to 20 kHz.

(74) The inkjet printhead according to the disclosure may include the plurality of nozzles **131** arranged in a row direction and a column direction, forming a matrix. FIG. 7 shows an example that the nozzles **131** are arranged in a matrix having three rows and seven columns, but the arrangement of the nozzles **131** is not limited to this example.

(75) Under the nozzles **131**, the third electrode **152** and the fourth electrode **153** surrounded with the insulating layer **151** may be arranged. In this case, the plurality of third electrodes **152** arranged in one of the row and column directions may be electrically connected in a straight line. FIG. 7 shows that seven third-electrodes **152** are electrically connected in the row direction, and thus three rows a, b and c of third electrodes **152** are connected to the second voltage controller **155**.

Likewise, the fourth electrodes **153** arranged in one of the row and column directions may be electrically connected in a straight line. In this case, the third electrodes **152** and the fourth electrodes **153** are arranged being orthogonal to each other. When the third electrodes **153** are electrically connected in the row direction, the fourth electrode **153** may be electrically connected in the column direction. In FIG. 7, three fourth electrodes are electrically connected in the column direction, and thus seven columns **1, 2, 3, 4, 5, 6** and **7** of fourth electrodes **153** are connected to the third voltage controller **157**.

(76) If the plurality of third electrodes **152** and the plurality of fourth electrodes **153** are individually connected to the second voltage controller **155** and the third voltage controller **157**, a voltage connection structure (circuit) may be complicated. The complicated voltage connection structure increases the distance between the nozzles **131**, so that the plurality of nozzles **131** cannot be compactly arranged. Furthermore, when high voltage is individually applied to the third electrodes **152** and the fourth electrodes **153**, the voltage controllers **155** and **157** may be burdened in terms of their capacities. On the other hand, according to the disclosure, the plurality of third electrodes **152** and the plurality of fourth electrodes **153** are electrically connected in the row direction or the column direction, thereby simplifying the voltage connection structure and lessening the burdens of the voltage controllers **155** and **157** in terms of capacity.

(77) To discharge droplets through some nozzles **131** among the plurality of arranged nozzles **131**, voltage may be applied to lines connected to the electrodes **152** and **153** corresponding to these nozzles **131**. For example, to discharge the droplets through only the nozzles **131** marked in black among the plurality of nozzles **131** as shown in FIG. 8, the second voltage controller **155** and the third voltage controller **157** may apply voltage to the lines b, c, **1, 2** and **3** electrically connected to the third electrodes **152** and the fourth electrodes **153** corresponding to the marked nozzles **131**. In other words, the lines b, c, **1, 2** and **3** may be on, but the lines a, **4, 5, 6** and **7** may be off. In this case, only the piezoelectric actuators **140** corresponding to the nozzles **131** marked in black are operated, and the piezoelectric actuators **140** corresponding to the other nozzles **131** are not operated, so that the droplets can be controlled not to be discharged from the other nozzles **131** (marked with a dotted line) to which voltage is applied while the voltage is applied to the lines b, c, **1, 2** and **3**.

(78) Below, an inkjet printhead according to a third embodiment of the disclosure will be described with reference to FIGS. **9** and **10**.

(79) FIG. **9** is a cross-sectional view of the inkjet printhead according to the third embodiment of the disclosure, and FIG. **10** is a view showing an operation of a membrane based on a piezoelectric body and a fifth electrode of FIG. **9**.

(80) In terms of describing the third embodiment, descriptions will be made focusing on differences from the foregoing embodiments.

(81) Even in this embodiment, the inkjet printhead has a three-layered structure of the chamber layer **110**, the channel layer **120** and the nozzle layer **130**.

(82) As compared with the first embodiment, this embodiment may additionally include a fifth electrode **171** surrounded with an insulating layer **170** and disposed on the channel layer **120**. Therefore, the fifth electrode **171** surrounded with the insulating layer **170** may be disposed between the chamber layer **110** and the channel layer **120**.

(83) A fourth voltage controller **175** applies voltage to the fifth electrode **171**.

(84) While the plurality of third electrodes **152** are individually and respectively disposed for the nozzles **131**, the fifth electrode **171** in this embodiment may form a common electrode with regard to the plurality of nozzles **131**. The fifth electrode **171** reinforces induced electric force by the third electrode **152**, thereby assisting the ink in forming the meniscus. Actual triggering of droplet detachment may be caused by potential difference from the electrostatic force applied to the third electrode **152**. With the fifth electrode **171**, it is possible to decrease the minimum extraction voltage of the third electrode **152**.

(85) The fifth electrode **171** may be used for generating potential difference from the electrostatic force of the third electrode **152**. Optionally, the fifth electrode **171** may receive voltage having the opposite polarity to that of the voltage applied to the third electrode **152**, or may serve as the ground for the ink.

(86) Further, the fifth electrode **171**, together with the first electrode **141** of the piezoelectric actuator **140**, generates the electrostatic force so that the membrane **115** can oscillate based on the electrostatic force. The electrostatic force between the first electrode **141** and the fifth electrode **171** may be added to the driving force of the piezoelectric body **142**. Therefore, it is possible to reinforce the oscillation of the membrane **115** and precisely control the discharging force based on the oscillation of the membrane **115**.

(87) Below, an inkjet printhead according to a fourth embodiment of the disclosure will be described with reference to FIG. **11**.

(88) FIG. **11** is a cross-sectional view of the inkjet printhead according to the fourth embodiment of the disclosure.

(89) In terms of describing the fourth embodiment, descriptions will be made focusing on differences from the foregoing embodiments.

(90) Even in this embodiment, the inkjet printhead has a three-layered structure of the chamber layer **110**, the channel layer **120** and the nozzle layer **130**.

(91) In this embodiment, the fourth electrode **153** according to the second embodiment may be additionally provided. As described above, the third electrode **152** coated with the insulating layer **151** is disposed beneath the nozzle layer **130**. The third electrode **152** and the fourth electrode **153** are all disposed inside the insulating layer **151** beneath the nozzle layer **130**, in which the fourth electrode **153** is disposed below and spaced apart from the third electrode **152**, and the horizontal distance between the fourth electrode **153** and the nozzle **131** is designed to be longer than the horizontal distance between the third electrode **152** and the nozzle **131**.

(92) The hydrophobic coating layer **160** coated with a hydrophobic material may be formed beneath the insulating layer **151** surrounding the third electrode **152** and the fourth electrode **153**.

(93) The third voltage controller **157** applies voltage to the fourth electrode **153**.

(94) When a droplet is formed and discharged by induced electrostatic force of the third electrode **152** and the pressure wave caused by the piezoelectric body **142** as the first voltage controller **145** applies voltage to the electrodes **141** and **143** of the piezoelectric actuator **140** and the second voltage controller **155** applies voltage to the third electrode **152**, the third voltage controller **157** applies voltage for forming the electrostatic force to the fourth electrode **153**, thereby narrowing a discharging angle of droplets and reinforcing a discharging speed. In this case, the voltage applied

to the fourth electrode **153** is synchronized to have the same phase as the voltage applied to the third electrode **152**.

(95) The vertical distance between the third electrode **152** and the end of the nozzle **131** is much shorter than the distance between the printhead and the printing medium. Therefore, a strong electric field is locally formed around the nozzle **131** by not the distance between the printhead and the printing medium but the third electrode **152**.

(96) Further, in this embodiment, the fifth electrode **171** according to the third embodiment may be additionally provided.

(97) In more detail, the fifth electrode **171** coated with the insulating layer **170** may be additionally disposed on the channel layer **120**. Therefore, the insulating layer **170** may be disposed between the chamber layer **110** and the channel layer **120**.

(98) The fourth voltage controller **175** applies voltage to the fifth electrode **171**.

(99) Thus, according to this embodiment, the force of the pressure wave based on the oscillation of the piezoelectric body **142** by the first voltage controller **145** and the piezoelectric actuator **140**, the force of induced electrostatic force by the second voltage controller **155** and the third electrode **152**, the auxiliary force of the induced electrostatic force by the third voltage controller **157** and the fourth electrode **153**, and the force of the pressure wave based on the induced electrostatic force or the electrostatic force by the fourth voltage controller **175** and the fifth electrode **171** are combined to produce the droplet at the end of the nozzle **131** and discharge the droplet while precisely controlling the discharging speed and direction of the droplet.

(100) Below, a method of manufacturing an inkjet printhead according to an embodiment of the disclosure will be described with reference to FIGS. **12** to **15**.

(101) FIG. **12** is a flowchart showing a method of manufacturing an inkjet printhead according to an embodiment of the disclosure, FIG. **13** is a view showing a method of manufacturing a chamber layer, FIG. **14** is a view showing a method of manufacturing a channel layer, and FIG. is a view showing a method of manufacturing a nozzle layer.

(102) The method of manufacturing the inkjet printhead according to an embodiment of the disclosure includes the steps of manufacturing the chamber layer **110** (or a first layer), the channel layer **120** (or a second layer), and the nozzle layer **130** (or a third layer) (**S10**); joining the chamber layer **110**, the channel layer **120**, and the nozzle layer **130** (**S20**); and forming the piezoelectric body **142** and the second electrode **143** of the piezoelectric actuator **140** on the chamber layer **110** (**S30**).

(103) The step of manufacturing the chamber layer **110** is illustrated in FIG. **13**.

(104) The chamber layer **110** may be manufactured with a silicon-on-insulator (all) wafer S. The thickness of the SOI wafer S is adjusted to a designed value by a chemical mechanical planarization (CMP) process, and then Si is etched by a photo process and an etching process to process the plurality of chambers **112**, thereby forming the membrane **115** on each chamber **112**. Further, the inlet **111** is formed to penetrate the SOI wafer S. Next, a lower electrode of the piezoelectric actuator **140**, i.e., the first electrode **141** is formed on the chamber layer **110** by using a shadow mask and a sputter.

(105) The step of manufacturing the channel layer **120** is illustrated in FIG. **14**.

(106) The channel layer **120** may be manufactured with either an Si wafer S or a glass wafer S. In the case of the Si wafer S, the photo process and the etching process may be used like the method of manufacturing the chamber layer **110** when the manifold **121** and the plurality of nozzle channels **122** are formed. In this case, the channel layer **120** is manufactured to have a thickness of 400 μm to 500 μm . In the case of using the glass wafer S, etching may be performed using a sand blast for chemical or physical etching. The induced electrostatic force may be reinforced by adding the fifth electrode **171** surrounded with the insulating layer **170** onto the channel layer **120**, and the oscillation of the membrane **115** may be reinforced by the electrostatic force.

(107) In this case, the fifth electrode **171** may be insulated by a dielectric material. After depositing

the insulating layer **170** of SiO.sub.2 on the wafer of the channel layer **120**, the shadow mask is aligned and the sputter is used to form a metal layer (i.e., the fifth electrode **171**). Next, the metal layer is protected with the insulating layer **170** of SiO.sub.2, so as not to be in direct contact with the ink. In the case of the glass wafer S, the shadow mask and the sputter are directly used to deposit the metal layer because the glass wafer S itself is an insulating material, and then the metal layer is protected by depositing the insulating layer **170** on the entire surface of the metal layer so as not to be in contact with ink.

(108) In the foregoing first and second embodiments, the insulating layer **170** and the fifth electrode **171** may be omitted.

(109) The step of manufacturing the nozzle layer **130** is illustrated in FIG. **15**.

(110) The method of manufacturing the nozzle layer **130** is similar to the method of manufacturing the chamber layer **110**. The nozzle layer **130** may be manufactured with a Si wafer S. For more precis manufacturing, an SOI wafer S may be used. A glass wafer may also be used. The nozzle layer **130** may be manufactured to have a thickness of 200 μm to 400 μm . On the top of the wafer S, the photo process and the etching process are used to form a channel for the nozzle **131**. Then, the outlet **154** is formed by applying the photo process and the etching process to the bottom of the wafer S. The size of nozzle **131** may be varied depending on the size of droplets to be discharged. To discharge droplets of several picoliters or less, i.e., ultrafine droplets. The nozzle **131** may have a diameter of 10 μm or less. To discharge large droplets, the nozzle **131** may have a diameter of 50 μm or more. The bottom of the nozzle layer **130** may be protected with the insulating layer **151**. For example, SiO.sub.2 may be deposited on the bottom of the nozzle layer **130**. In this case, the insulating layer **151** may be formed encompassing the inside of the outlet **154** and the inside of the nozzle **131**.

(111) The third electrode **152** for forming the induced electrostatic force may be formed beneath the nozzle layer **130**. In this case, the third electrode **152** is surrounded with the insulating layer **151** so as not to be in contact with ink. As described above, the third electrodes **152** in this case may be integrally formed for the plurality of nozzles **131** or may be individually separated for the nozzles **131**.

(112) When the second voltage controller **155** applies voltage to the third electrode **152**, charges are induced in the ink inside the nozzle **131** and thus droplets are discharged through the opening of the nozzle **131** by the electric field around the nozzle **131**. In this case, DC voltage, DC pulse voltage, or AC voltage may be applied to the third electrode **152**. The droplets may be discharged by a process that charges induced in the ink are continuously dissipated and induced again, and this process may be electrically construed as the flow of the induced current.

(113) When the piezoelectric body **142** is operated by the continuous DC voltage and the pulse applied to the piezoelectric body **142** to discharge droplets, the droplets hit the printing medium while dissipating the induced charges, and it is therefore possible to discharge the droplets. In this case, the DC voltage may be lower than or equal to an onset voltage of when the droplet is discharged by only the electrostatic force.

(114) Even when the droplets are continuously discharged by applying the DC pulse to the third electrode **152** and operating the piezoelectric body **142**, the generation of induced charges based on the generation and discharge of the droplets may be continued.

(115) In terms of the induced charges, the most preferable voltage is the AC voltage. When positive and negative charges are alternately generated, it is easier to continuously generate the induced charges. However, even when the DC voltage is applied, continuous ink operation is possible as described above without any problems because the induced charges are dissipated and new charges are induced in the ink as the droplets are discharged.

(116) The third electrode **152** is formed to surround the nozzle **131** by depositing the electrode layer on the insulating layer **151** formed beneath the nozzle layer **130**, and then applying the photo process and the etching process to the deposited electrode layer. In this case, the third electrode **152**

may have various shapes such as a circular shape, a horseshoe shape, a quadrangular shape, a diamond shape, etc. as long as it can surround all or some of the nozzles **131**. When it is assumed that the third electrode **152** has the circular shape, the inner diameter of the third electrode **152** may be larger than or equal to the outer diameter of the nozzle **131**. By depositing the insulating layer **151** on the processed third electrode **152**, the third electrode **152** may be disposed to be surrounded with the insulating layer **151**. SiO₂, SiN_x, etc. may be deposited as the insulating layer, and polyimide or the like polymer layer may be deposited.

(117) When the nozzle layer **130** is made of an insulating material, i.e., the glass wafer, the third electrode **152** may be directly formed beneath the nozzle layer **130** without the insulating layer, and the insulating layer **151** may be formed beneath the third electrode **152**, as shown in FIG. **15**. In this case, the insulating layer **151** may be formed even inside the outlet **154** and the nozzle **131**.

(118) Further, the fourth electrode **153** may be additionally formed below the third electrode **152** surrounded with the insulating layer **151**, and the insulating layer **151** may be applied below the fourth electrode **153**.

(119) Next, to have a water repellent effect, the hydrophobic coating layer **160** made of hydrophobic material based on a fluoropolymer may be formed beneath the insulating layer **151**. The water repellent layer may be formed by a spray coating method, an e-beam evaporator, a sputter or the like vacuum deposition.

(120) The hydrophobic coating layer **160** may be formed inward up to a predetermined depth from the end of the nozzle **131**. When the hydrophobic coating layer **160** is formed, the hydrophobic coating layer **160** is coated while air is sprayed through the nozzle **131**. In this case, the depth, up to which the hydrophobic coating layer **160** is coated on the inside of the nozzle **131**, may be controlled according to the speeds of sprayed air.

(121) As described above, the chamber layer **110**, the channel layer **120**, the nozzle layer **130** are individually manufactured, and then the chamber layer **110**, the channel layer **120**, and the nozzle layer **130** are jointed together. First, the chamber layer **110** and the channel layer **120** are joined by anodic bonding, and then the nozzle layer **130** is joined beneath the channel layer **120** by anodic bonding.

(122) When both the two layers to be joined are manufactured with the silicon wafer, they may be joined by direct bonding. Further, when the channel layer **120** is manufactured with the glass wafer, the anodic bonding may be used. When the channel layer **120** is manufactured with the Si wafer, Si direct bonding may be used.

(123) Next, the lower electrode (i.e., the first electrode **141**) of the piezoelectric actuator **140** is formed on the completed device. The piezoelectric body **142** is formed on the first electrode **141**, and the second electrode **143** is deposited on the piezoelectric body **142**, thereby completing the piezoelectric actuator **140**. The piezoelectric body **142** may be fixed as a bulk piezoelectric body **142** onto the membrane **115** of the chamber layer **110** by bonding. Alternatively, the sputter may be used to deposit the piezoelectric body **142**, and then the photo and etching processes may be performed. Alternatively, screen printing or the like printing method may be used to apply a material of the piezoelectric body **142**, thereby forming the layer of the piezoelectric body **142**.

(124) In the foregoing inkjet printhead according to the disclosure and the method of manufacturing the same, electrostatic force induced in a meniscus, which is formed at the end of the nozzle by the pressure wave based on the operation of the piezoelectric body, is used to generate droplets, and it is thus possible to generate stronger discharging force than that of the inkjet printhead based on the operation of the piezoelectric body, thereby having advantages in that ink having a viscosity of 30 cP higher than 10 cP, which is the viscosity the ink dischargeable by the conventional inkjet printhead has, is dischargeable and even ink having a viscosity of 100 cP or higher is also dischargeable.

(125) Further, discharging power is so strong that droplets can be discharged even when the diameter of the nozzle is smaller than that of the conventional inkjet printhead, thereby having

advantages in that ink droplets can be discharged at a femtoliter level (for reference, it is difficult for a conventional inkjet printhead to stably discharge droplets of not more than 1 picoliter (i.e., a diameter of 12.4 μm)).

(126) Further, the nozzles of the multi-nozzle are controlled to be different in the level of the high voltage applied to the electrode for electrohydrodynamic jetting while an electric signal for driving the piezoelectric body is maintained constant, thereby having advantages in that the droplets discharged through the nozzles can be uniform in size.

(127) Further, the fourth electrode is additionally disposed beneath the third electrode, thereby having advantageous in the discharging force and direction are more precisely controlled.

(128) Further, the fifth electrode is additionally disposed between the channel layer and the chamber layer, thereby having advantageous in that the induced electrostatic force of the third electrode is reinforced or the oscillation of the membrane is strengthened by the electrostatic force acting between the first electrode and the fifth electrode.

(129) Although detailed embodiments of a fluidic lens with a variable focal length according to the disclosure have been described, the disclosure is not limited to such detailed embodiments. Various changes and modifications can be made by a person having ordinary knowledge in the art without departing from the spirit and scope of the invention defined in the appended claims.

Claims

1. A inkjet printhead, comprising: a first layer comprising an inlet formed to penetrate a substrate and introduce ink therein, and a plurality of membranes; a second layer disposed beneath the first layer, and comprising a manifold formed to penetrate a substrate or to be recessed on a top of the substrate to communicate with the inlet, and a plurality of nozzle channels formed to penetrate the substrate below the membrane and allow the ink transferred from the manifold to flow therein; a third layer disposed beneath the second layer, and comprising a plurality of nozzles formed in a substrate and communicating the plurality of nozzle channels; a piezoelectric actuator formed on the first layer formed with the membrane, and comprising a lower first electrode, a piezoelectric body on the first electrode, and a second electrode on the piezoelectric body; a first voltage controller configured to oscillate the membrane by applying a pulse voltage to the first electrode and the second electrode; a third electrode disposed beneath the third layer, formed around each nozzle, and surrounded with an insulator; a second voltage controller configured to discharge droplets of the ink based on induced electric force by applying voltage to the third electrode; a fifth electrode disposed between the first layer and the second layer and surrounded with an insulating layer; and a fourth voltage controller configured to apply voltage to the fifth electrode.
2. The inkjet printhead of claim 1, wherein the first layer comprises a plurality of chambers recessed inward from a bottom thereof to store the ink and formed with the membrane above a top thereof.
3. The inkjet printhead of claim 2, wherein the chamber communicates with the manifold at a first side and communicates with the nozzle channel at a second side.
4. The inkjet printhead of claim 1, further comprising a hydrophobic coating layer disposed beneath an insulating layer formed the insulator and coated with a hydrophobic material.
5. The inkjet printhead of claim 4, wherein the hydrophobic coating layer is coated from an end of the nozzle to an inside of the nozzle.
6. The inkjet printhead of claim 1, wherein voltage applied by the first voltage controller is synchronized with voltage applied by the second voltage controller.
7. The inkjet printhead of claim 6, wherein the second voltage controller applies voltage to the third electrode to discharge a droplet when a meniscus is formed at an end of the nozzle as the first voltage controller applies a pulse voltage between the first electrode and the second electrode to make the piezoelectric actuator oscillate the membrane, and the second voltage controller applies a

- voltage having an opposite polarity to a discharging voltage to the third electrode or applies a voltage of OV to the third electrode after the discharged droplet passes the third electrode.
8. The inkjet printhead of claim 1, further comprising: a fourth electrode disposed beneath the third electrode, surrounded with an insulating layer, and disposed encompassing an outlet having a larger diameter than an opening of the nozzle; and a third voltage controller configured to apply voltage to the fourth electrode.
9. The inkjet printhead of claim 8, wherein voltage applied by the second voltage controller is synchronized with voltage applied by the third voltage controller.
10. The inkjet printhead of claim 8, wherein a horizontal distance between the fourth electrode and the nozzle is longer than a horizontal distance between the third electrode and the nozzle.
11. The inkjet printhead of claim 8, wherein the plurality of nozzles are arranged in a matrix, and the third electrodes arranged in one of a row direction and a column direction are electrically connected to simultaneously receive voltage from the second voltage controller, and the fourth electrodes arranged in the other one of the row direction and the column direction are electrically connected to simultaneously receive voltage from the third voltage controller.
12. The inkjet printhead of claim 1, wherein the first voltage controller applies the same pulse voltage to the piezoelectric actuators respectively corresponding to the membranes, and the second voltage controller applies different voltages to the third electrodes according to the nozzles to make droplets discharged from the nozzles be uniform in size.
13. The inkjet printhead of claim 1, wherein the fourth voltage controller applies voltage, which has an opposite polarity to the voltage of the second voltage controller, to the fifth electrode, or serves as the ground.
14. The inkjet printhead of claim 1, wherein the pulse voltage applied by the first voltage controller is synchronized with the pulse voltage applied by the fourth voltage controller, so that electrostatic force based on potential difference between the first electrode and the fifth electrode can reinforce the oscillation of the membrane.
15. The inkjet printhead of claim 1, wherein the third electrodes are integrally formed for the plurality of nozzles.
16. A method of manufacturing an inkjet printhead, comprising: manufacturing a first layer comprising an inlet formed to penetrate a substrate and introduce ink therein, and a plurality of membranes on which a first electrode of a piezoelectric actuator is formed, a second layer comprising a manifold formed to penetrate a substrate or to be recessed on a top of the substrate to communicate with the inlet, and a plurality of nozzle channels formed to penetrate the substrate below the membrane and allow the ink transferred from the manifold to flow therein, and a third layer comprising a plurality of nozzles formed in a substrate and communicating the plurality of nozzle channels, and a third electrode disposed beneath the third layer, formed around each nozzle, and surrounded with an insulator; joining the first layer, the second layer, and the third layer in sequence; forming a piezoelectric body and a second electrode for a piezoelectric actuator on the first electrode; and forming a fifth electrode disposed on the second layer and surrounded with an insulator when the third layer is manufactured.
17. The method of claim 16 the first layer comprises a plurality of chambers recessed inward from a bottom thereof to store the ink and formed with the membrane above a top thereof.
18. The method of claim 16, further comprising coating a hydrophobic material on a lower layer of the insulator when the third layer is manufactured.
19. The method of claim 18, wherein a hydrophobic coating layer coated with the hydrophobic material is coated from an end of the nozzle to an inside of the nozzle.
20. The method of claim 16, further comprising forming a fourth electrode, disposed beneath the third electrode, surrounded with an insulator, and having a larger diameter than the third electrode when the third layer is manufactured.
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